

$$(V_i - V_{n1}) G_1 = (V_{n1} - V_{n2}) G_f$$

$$V_{n2} = \left(-\frac{G_1}{G_f} \right) V_i \rightarrow \text{理想因子}$$

$$(G_1 + G_i + G_f) V_{n1} - G_f V_{n2} = G_1 V_i \quad (1)$$

$$(A G_o - G_f) V_{n1} + (G_f + G_o + G_L) V_{n2} = 0 \quad (2)$$

$$V_{n1} = \frac{G_1 V_i + G_f V_{n2}}{(G_1 + G_i + G_f)}$$

$$\frac{(A G_o - G_f)(G_1 V_i + G_f V_{n2})}{(G_1 + G_i + G_f)} + (G_f + G_o + G_L) V_{n2} = 0$$

$$(A G_o - G_f)(G_1 V_i + G_f V_{n2}) + V_{n2}(G_f + G_o + G_L)(G_1 + G_i + G_f) = 0$$

$$(A G_o - G_f) G_1 V_i + (A G_o - G_f) G_f V_{n2} + V_{n2}(G_f + G_o + G_L)(G_1 + G_i + G_f) = 0$$

$$V_{n2} [(A G_o - G_f) G_f + (G_f + G_o + G_L)(G_1 + G_i + G_f)] = (G_f - A G_o) G_1 V_i$$

$$V_{n2} = \frac{(G_f - A G_o) G_1 V_i}{(A G_o - G_f) G_f + (G_f + G_o + G_L)(G_1 + G_i + G_f)}$$

$$V_{n2} = -\frac{G_1}{G_f} \frac{G_f V_i (A G_o - G_f)}{(A G_o - G_f) G_f + (G_f + G_o + G_L)(G_1 + G_i + G_f)}$$

$$A \rightarrow \infty \quad G_i \rightarrow \infty \quad G_o \rightarrow 0$$

$$V_{n2} = -\frac{G_1}{G_f} \frac{V_i A G_o}{(A G_o)} = -\frac{R_f}{R_1} V_i$$

$$\frac{V_o}{V_i} = -\frac{R_f}{R_1}$$

节点电压法(NCL)可以求解, 列KVL也可以